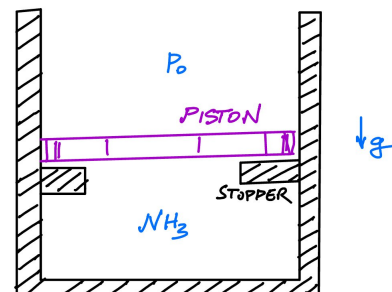


MEL2020 – Thermodynamics – Tutorial 04

1. A setup of piston cylinder contains 0.93 kg of ammonia at 298.05 K with a volume of 0.1 m³, as shown in the accompanying figure. Initially, the piston rests on some stops; its top surface is exposed to atmosphere. Atmospheric pressure can be assumed to be 100 kPa. Ammonia pressure of 1400 kPa is required to lift the piston. To what temperature must ammonia be heated to lift the piston? If the ammonia is heated to saturated vapor, find the final temperature, volume, and heat transferred to ammonia. Illustrate the process on a T-v diagram. Assume all processes to be quasi-equilibrium. [6 Marks]



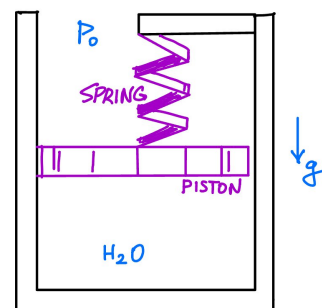
Saturation Table (Ammonia)

$P /$ MPa	$T^{\text{sat}} /$ K	$v^{\text{L}} /$ m ³ /kg	$v^{\text{V}} /$ m ³ /kg	$h^{\text{L}} /$ kJ/kg	$h^{\text{L,V}} /$ kJ/kg	$h^{\text{V}} /$ kJ/kg	$s^{\text{L}} /$ kJ/(kg · K)	$s^{\text{L,V}} /$ kJ/(kg · K)	$s^{\text{V}} /$ kJ/(kg · K)
1.0	298.05	0.001659	0.1285	460.32	1166.20	1626.52	1.8788	3.9128	5.7916
1.2	304.09	0.001684	0.1075	489.44	1140.33	1629.77	1.9744	3.7500	5.7244
* 1.4	309.40	0.001708	0.09230	515.36	1116.59	1631.95	2.0578	3.6089	5.6666

Superheated Table (Ammonia)

$P /$ MPa						$T /$ K				
$(T^{\text{sat}} / \text{K})$		sat	300	310	320	330	340	350	360	370
1.0	$v / \text{m}^3/\text{kg}$	0.1285	0.1299	0.1366	0.1430	0.1491	0.1551	0.1609	0.1666	0.1721
(298.0)	$h / \text{kJ/kg}$	1626.52	1632.60	1662.33	1690.39	1717.29	1743.38	1768.89	1794.00	1818.83
	$s / \text{kJ}/(\text{kg} \cdot \text{K})$	5.7916	5.8119	5.9095	5.9985	6.0813	6.1592	6.2332	6.3039	6.3719
1.4	$v / \text{m}^3/\text{kg}$	0.09230		0.09262	0.09785	0.1027	0.1074	0.1119	0.1162	0.1204
(309.4)	$h / \text{kJ/kg}$	1631.95		1633.97	1666.36	1696.48	1725.06	1752.56	1779.29	1805.47
	$s / \text{kJ}/(\text{kg} \cdot \text{K})$	5.6666		5.6732	5.7760	5.8687	5.9540	6.0338	6.1091	6.1808

2. An arrangement of piston/cylinder has a linear spring and the outside atmospheric pressure P_0 acting on the piston as shown in the accompanying figure. The cylinder contains water at 4 MPa and 650 K with a volume of 0.12 m³ (initial state). If the piston is at the very bottom of the cylinder, the spring exerts a force such that 300 kPa inside pressure is required to balance all forces. If the system now cools from its initial state until the pressure reaches 1 MPa, find the heat transfer for the cooling process. Assume the cooling process to be quasi-equilibrium. [9 Marks]



Saturation Table (Water)

$P /$ MPa	$T^{\text{sat}} /$ K	$v^{\text{L}} /$ m^3/kg	$v^{\text{V}} /$ m^3/kg	$h^{\text{L}} /$ kJ/kg	$h^{\text{LV}} /$ kJ/kg	$h^{\text{V}} /$ kJ/kg	$s^{\text{L}} /$ kJ/(kg · K)	$s^{\text{LV}} /$ kJ/(kg · K)	$s^{\text{V}} /$ kJ/(kg · K)
1.0	453.04	0.001127	0.1943	762.7	2014.4	2777.1	2.1384	4.4465	6.5850
1.2	461.11	0.001139	0.1632	798.5	1985.3	2783.8	2.2163	4.3054	6.5217
1.4	468.20	0.001149	0.1408	830.1	1958.8	2788.9	2.2839	4.1836	6.4675
1.6	474.53	0.001159	0.1237	858.6	1934.3	2792.9	2.3438	4.0762	6.4200
1.8	480.27	0.001168	0.1104	884.6	1911.4	2796.0	2.3978	3.9798	6.3776
2.0	485.53	0.001177	0.09958	908.6	1889.8	2798.4	2.4470	3.8921	6.3392
2.5	497.11	0.001197	0.07995	962.0	1840.1	2802.0	2.5544	3.7015	6.2560
3.0	507.01	0.001217	0.06666	1008.4	1794.9	2803.3	2.6456	3.5402	6.1858
4.0	523.51	0.001253	0.04978	1087.4	1713.5	2800.9	2.7967	3.2731	6.0697

Superheated Table (Water)

$P /$ MPa						$T /$ K				
$(T^{\text{sat}} /$ K)		sat	600	650	700	750	800	900	1000	1100
1.0	$v /$ m^3/kg	0.3668	0.3903	0.4137	0.4370	0.4603	0.4836	0.5069	0.5301	0.5533
453.0	$h /$ kJ/kg	3537.4	3647.2	3758.5	3871.3	3985.8	4101.8	4219.6	4339.0	4460.0
	$s /$ kJ/(kg · K)	7.8382	7.9714	8.0986	8.2206	8.3380	8.4512	8.5608	8.6669	8.7699
	$u /$ kJ/kg	3170.6	3257.0	3344.8	3434.3	3525.4	3618.2	3712.7	3808.9	3906.7
2.0	$v /$ m^3/kg	0.1821	0.1941	0.2060	0.2178	0.2296	0.2413	0.2530	0.2647	0.2764
485.5	$h /$ kJ/kg	3527.5	3638.8	3751.3	3865.1	3980.3	4097.0	4215.3	4335.2	4456.6
	$s /$ kJ/(kg · K)	7.5090	7.6439	7.7725	7.8956	8.0138	8.1277	8.2377	8.3442	8.4476
	$u /$ kJ/kg	3163.2	3250.6	3339.3	3429.5	3521.2	3614.4	3709.3	3805.7	3903.8
4.0	$v /$ m^3/kg	0.04978	0.06306	0.07027	0.07701	0.08350	0.08982	0.1021	0.1142	0.1229
523.5	$h /$ kJ/kg	2800.9	3034.3	3159.1	3277.3	3392.8	3507.3	3736.7	3969.3	4142.5
	$s /$ kJ/(kg · K)	6.0697	6.4877	6.6877	6.8629	7.0222	7.1700	7.4402	7.6852	7.8523
	$u /$ kJ/kg	2601.8	2782.0	2878.1	2969.3	3058.8	3148.0	3328.1	3512.5	3650.8